

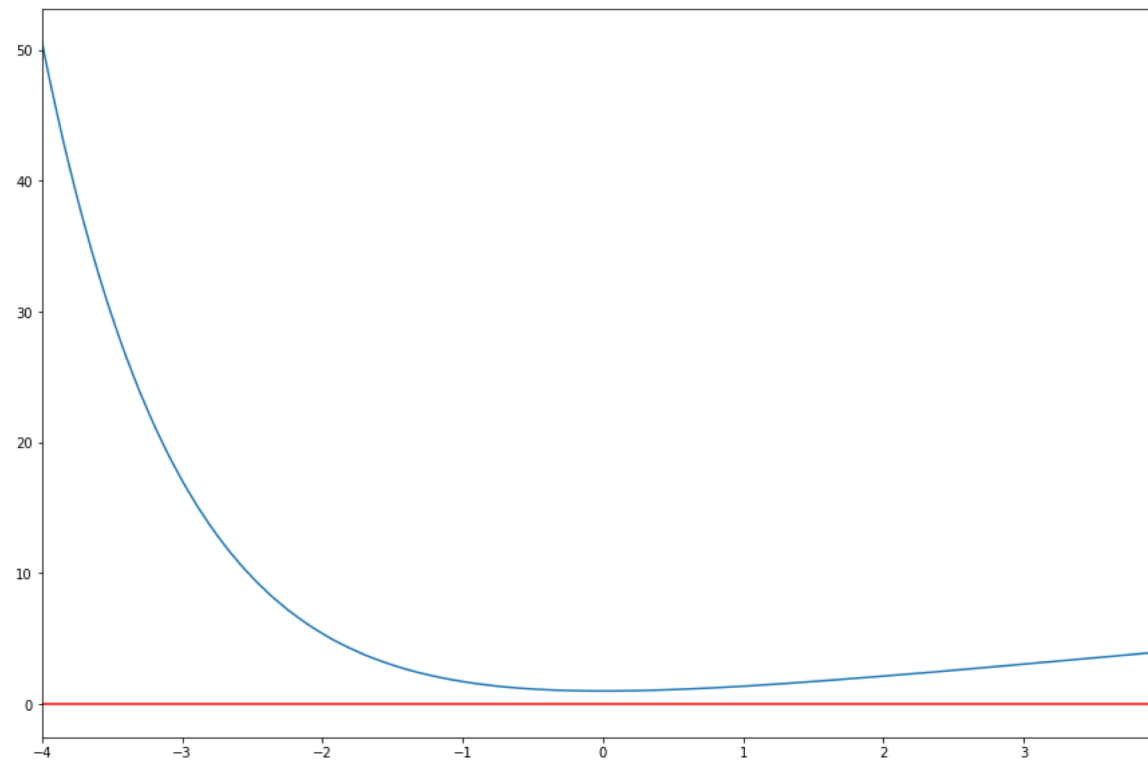
Solving non-linear equations

1D

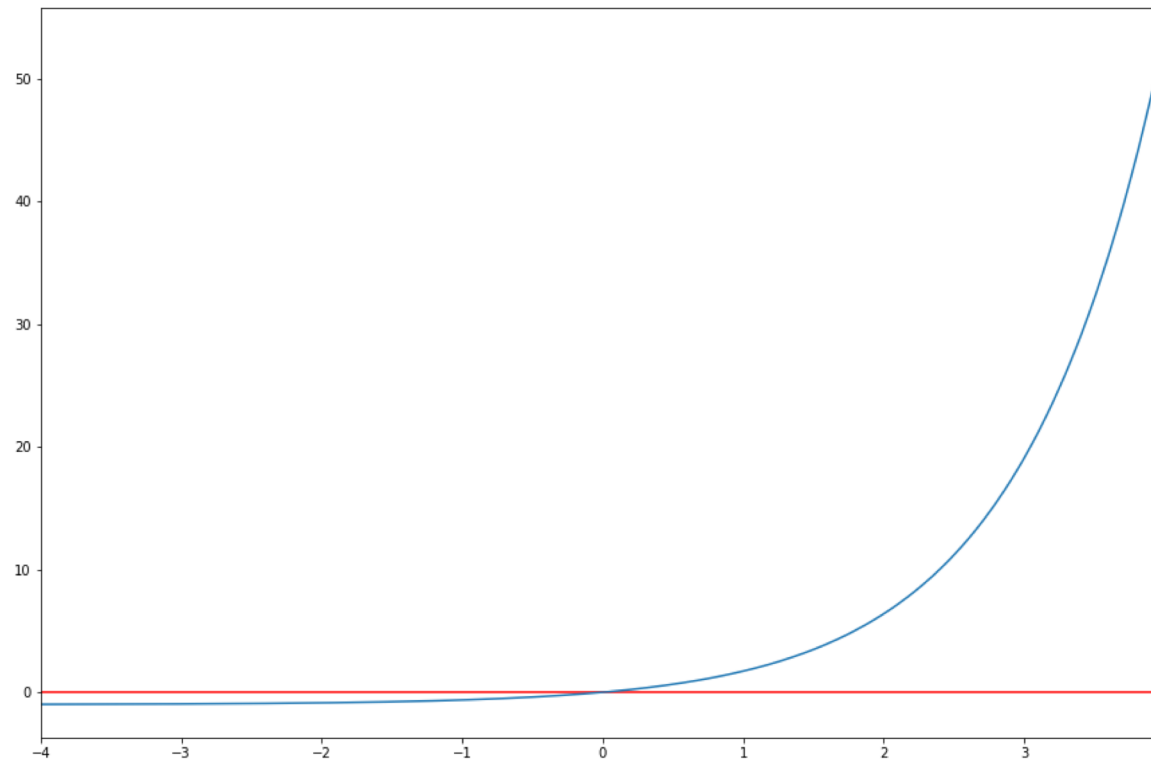
Motivation

- We want to find roots of nonlinear equations
 - Linear equations are easy 😊
- What we really want to do is find roots in systems of non-linear equations, but for now we look at a single equation

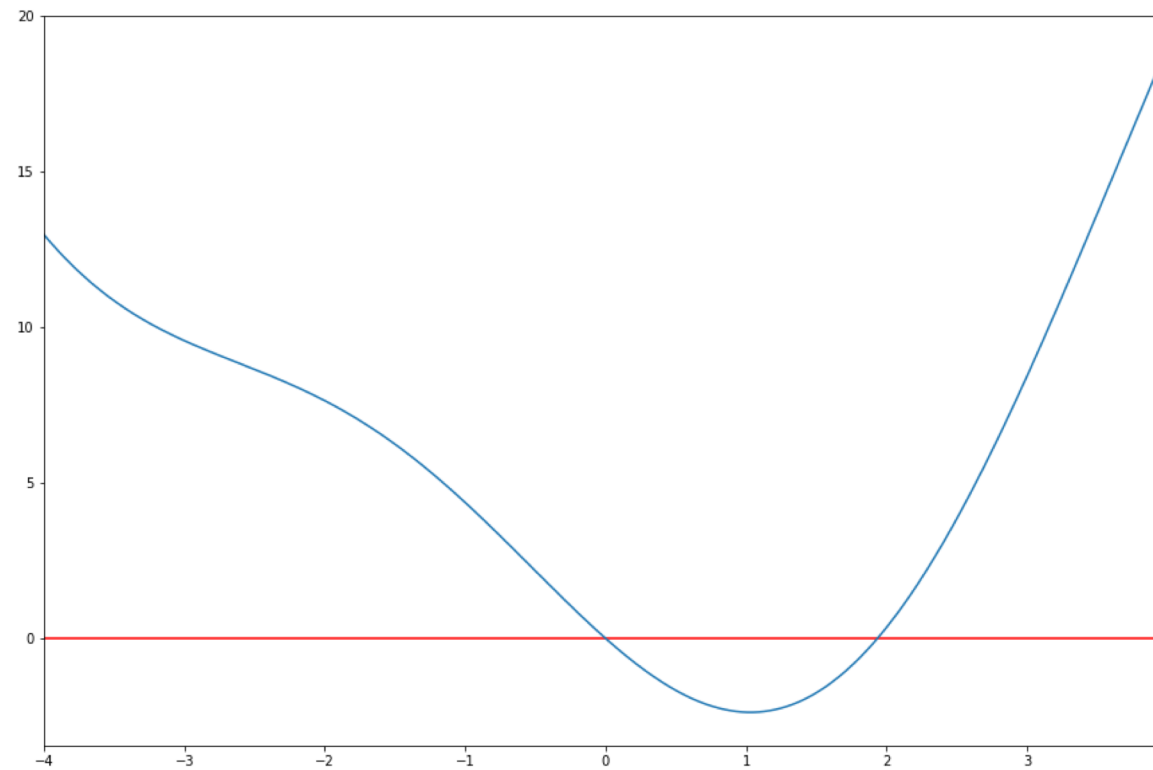
Example: $e^{-x} + x$



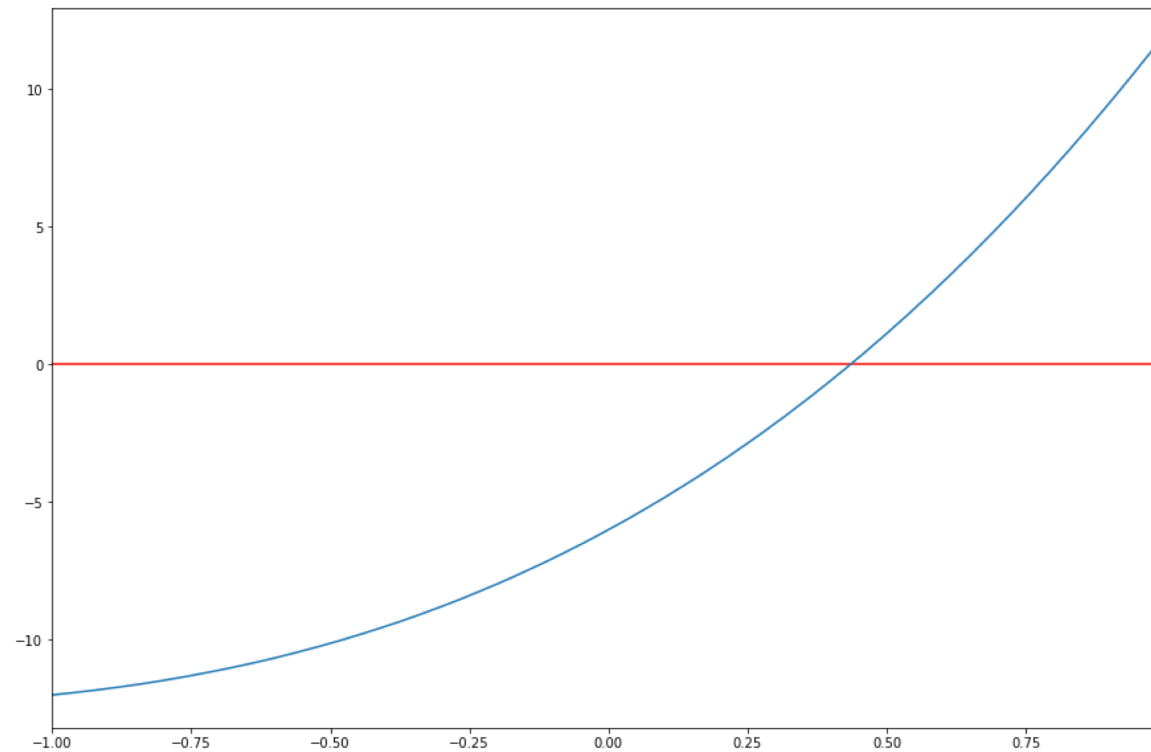
Example: $e^x - 1$



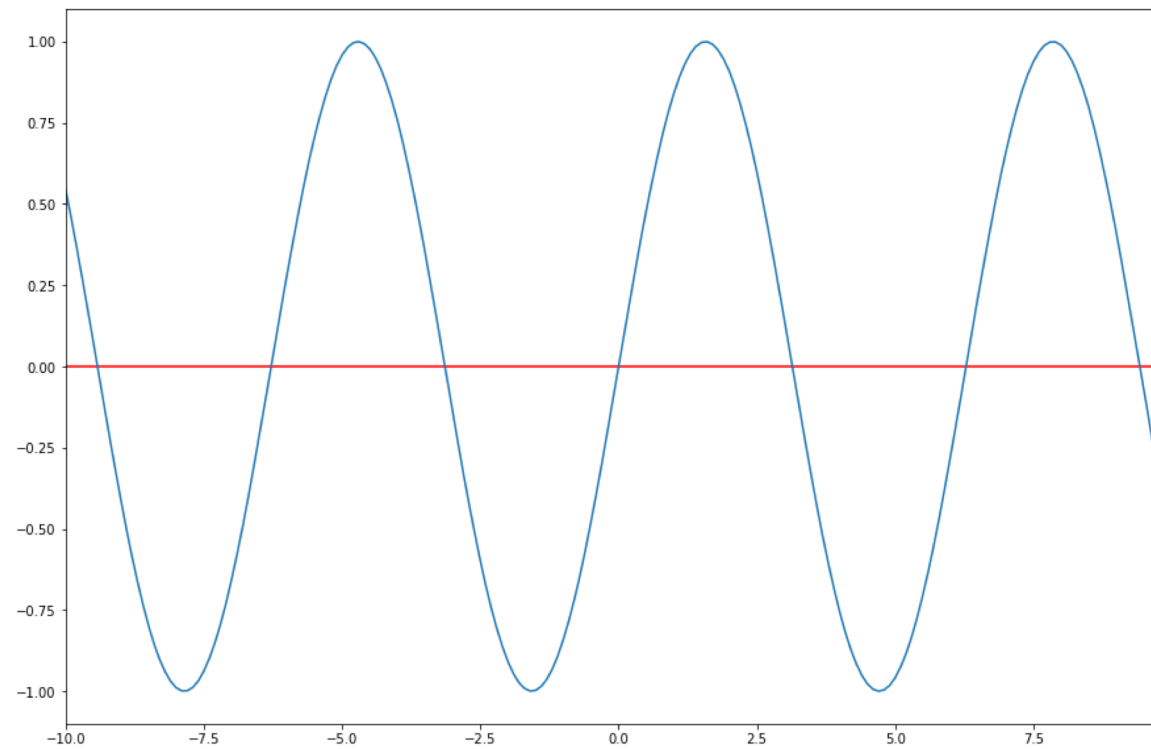
Example: $x^2 - 4\sin(x)$



Example: $x^3 + 6x^2 + 11x - 6$



Example: $\sin(x)$



Existence of a solution

Intermediate Value Theorem:

if f is continuous in $[a, b]$ and c is between $f(a)$ and $f(b)$ then there exists a value $x^* \in [a, b]$ such that $f(x^*) = c$.

So if $\text{sign}(f(a)) \neq \text{sign}(f(b))$ then the root must lie in $[a, b]$

Uniqueness

In 1D we only have the worry that a root may also be a minimum or a maximum. So if

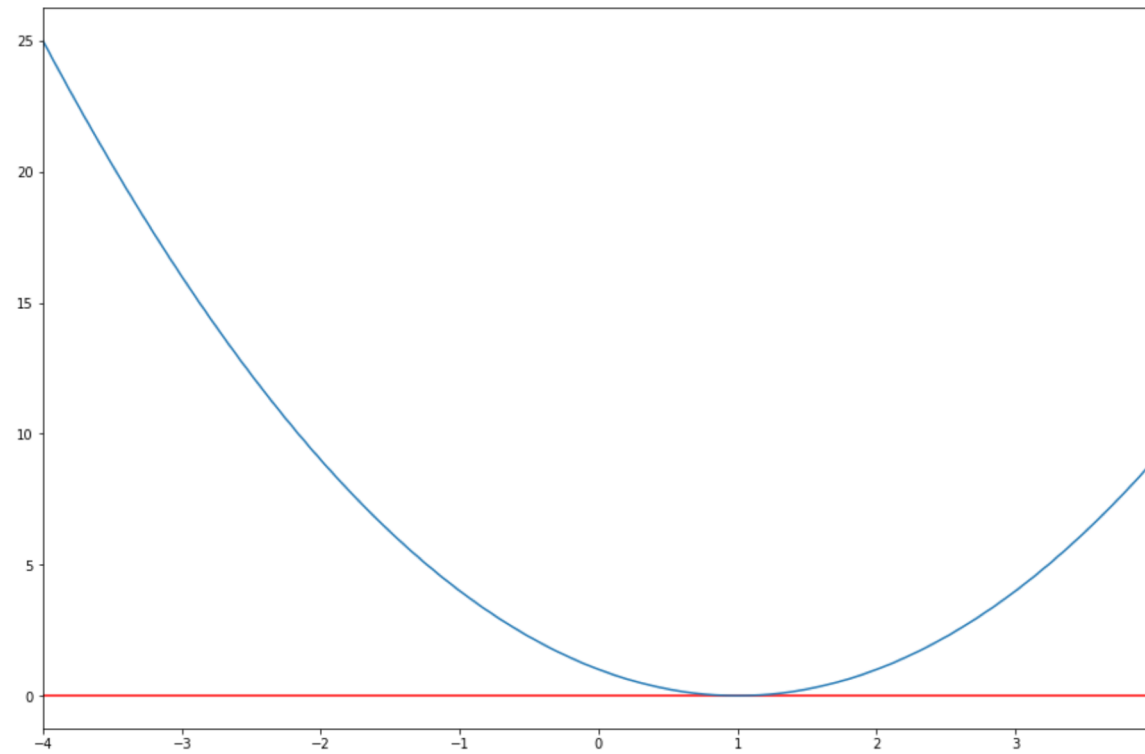
$$f(x) = 0 \wedge f'(x) \neq 0$$

we have a simple root.

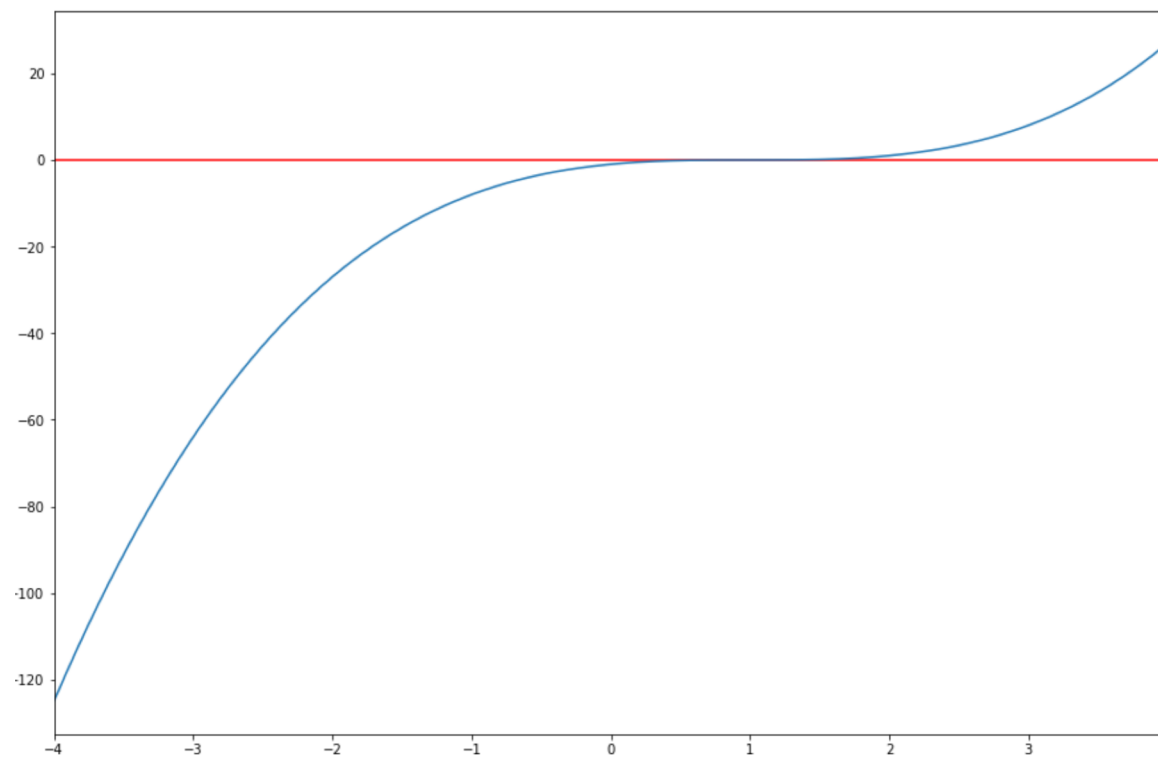
$$f(x) = 0 \wedge f'(x) = 0$$

we have a multiple root.

$$x^2 - 2x + 1$$



$$x^3 - 3x^2 + 3x - 1$$



Sensitivity

- How does the function behave around the root
 - Approaching rapidly
 - Approaching extremely slowly

Conditioning

We work with absolute conditioning numbers. As with linear equations we have that

$$c = \frac{1}{|f'(x^*)|}$$

this means if $|f(\hat{x})| \leq \epsilon$ error of $|\hat{x} - x^*|$ may be as large as $\frac{\epsilon}{|f'(x^*)|}$.

This means that in 1D a multiple root has an infinite conditioning number - we are obviously very sensitive numerically at this point.

Convergence Rates

$$\lim_{k \rightarrow \infty} \frac{||e_{k+1}||}{||e_k||^r} = C$$

$$e_k = x_k - x^*$$

Convergence Rates

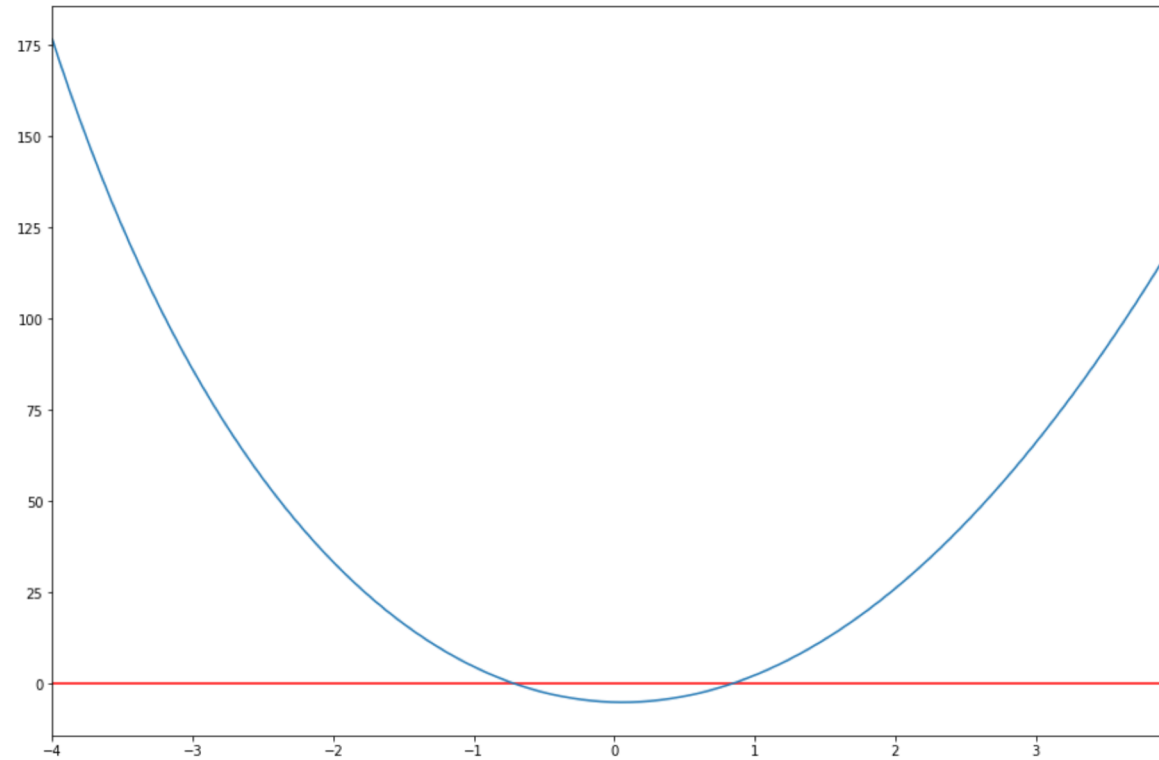
$$\lim_{k \rightarrow \infty} \frac{||e_{k+1}||}{||e_k||^r} = C$$

$$e_k = x_k - x^*$$

Stopping Criteria

$$\frac{||x_{k+1} - x_k||}{||x_k||}$$

Graphical Inspection



5 min exercise

- Find 3 functions that are well conditioned and 3 that are ill-conditioned
 - See if you can find functions from your field

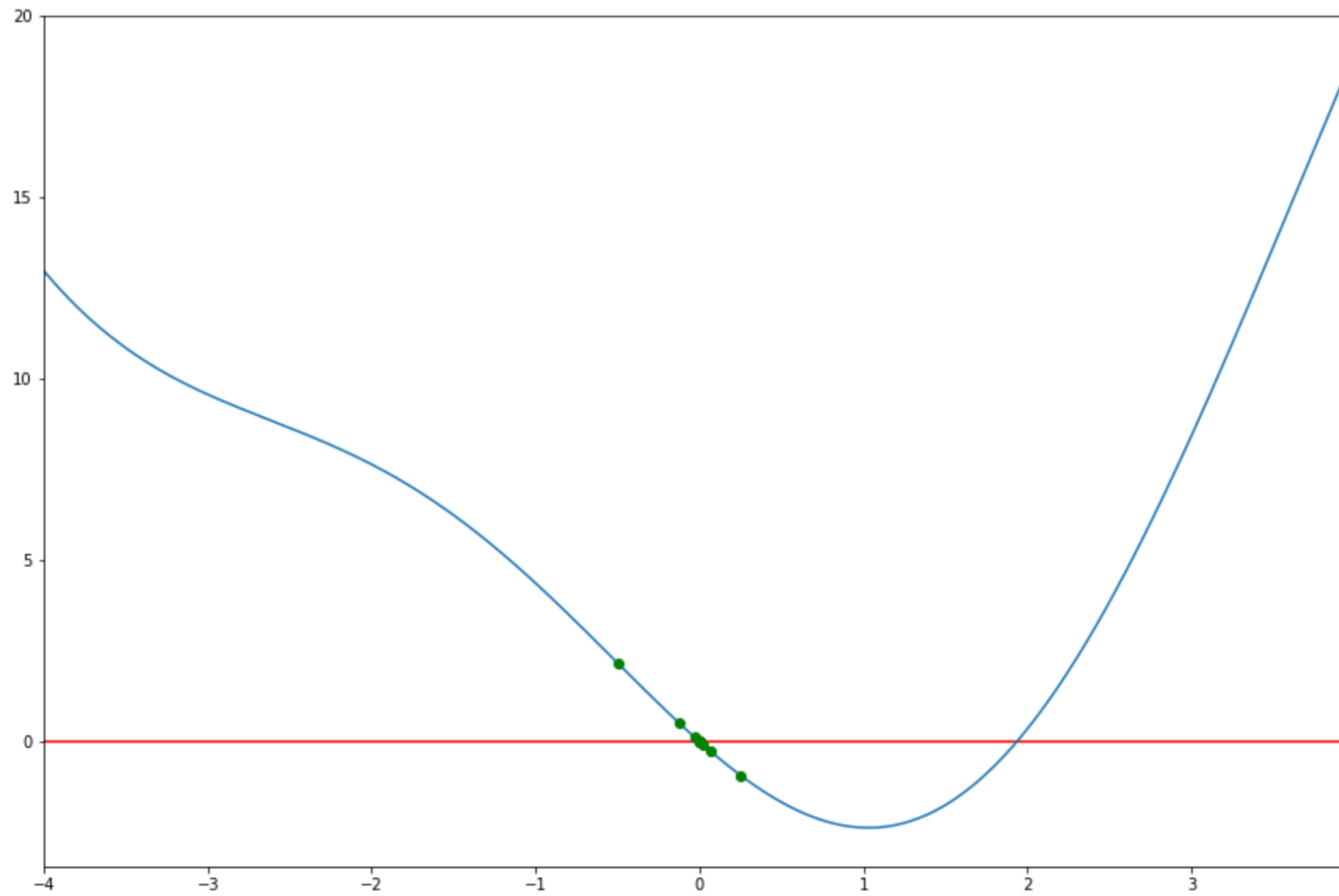
Bisection method

- Choose your initial brackets $[a, b]$ so that the root is in the bracket:
 - $\text{sign}(a) \neq \text{sign}(b)$
- Repeat
 - Find the sign of the mid of the bracket
 - Let the midpoint replace the bracket end with the same sign
- Tolerance

$$\lceil \log_2 \left(\frac{b-a}{\text{tol}} \right) \rceil$$

Code

Convergence

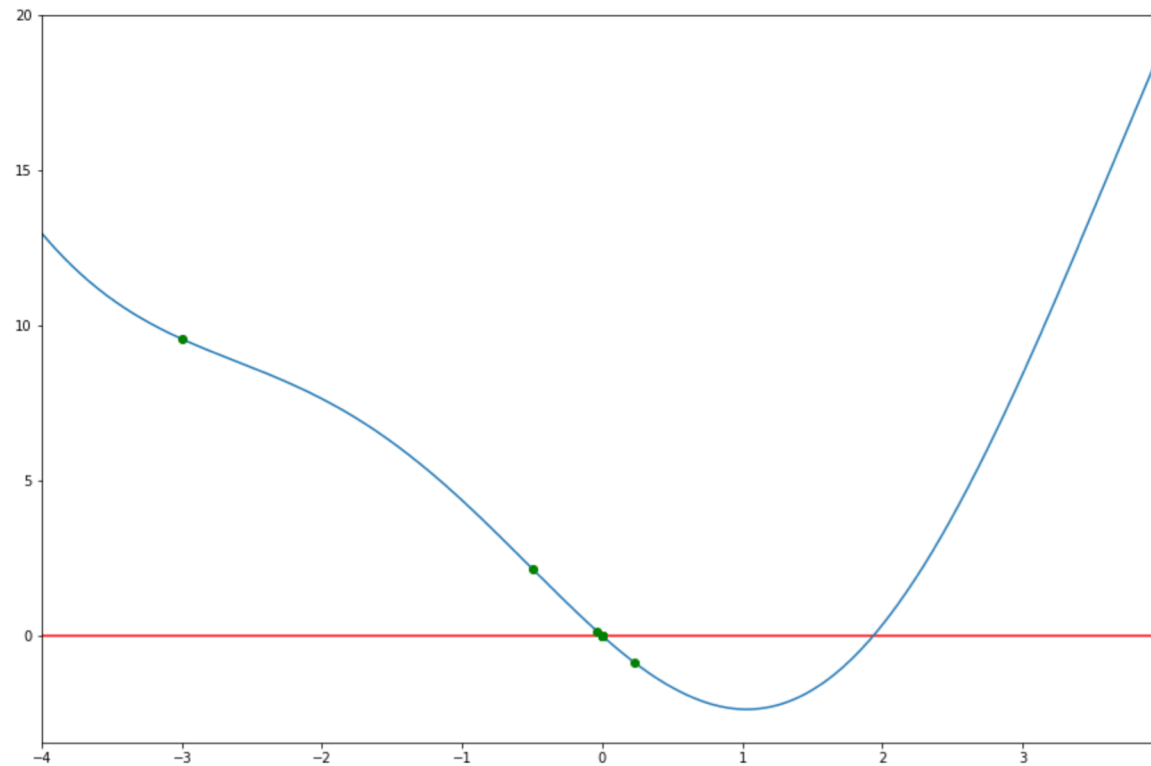


False Position

$$guess = b + \frac{((f(b)(a - b))}{(f(b) - f(a))})$$

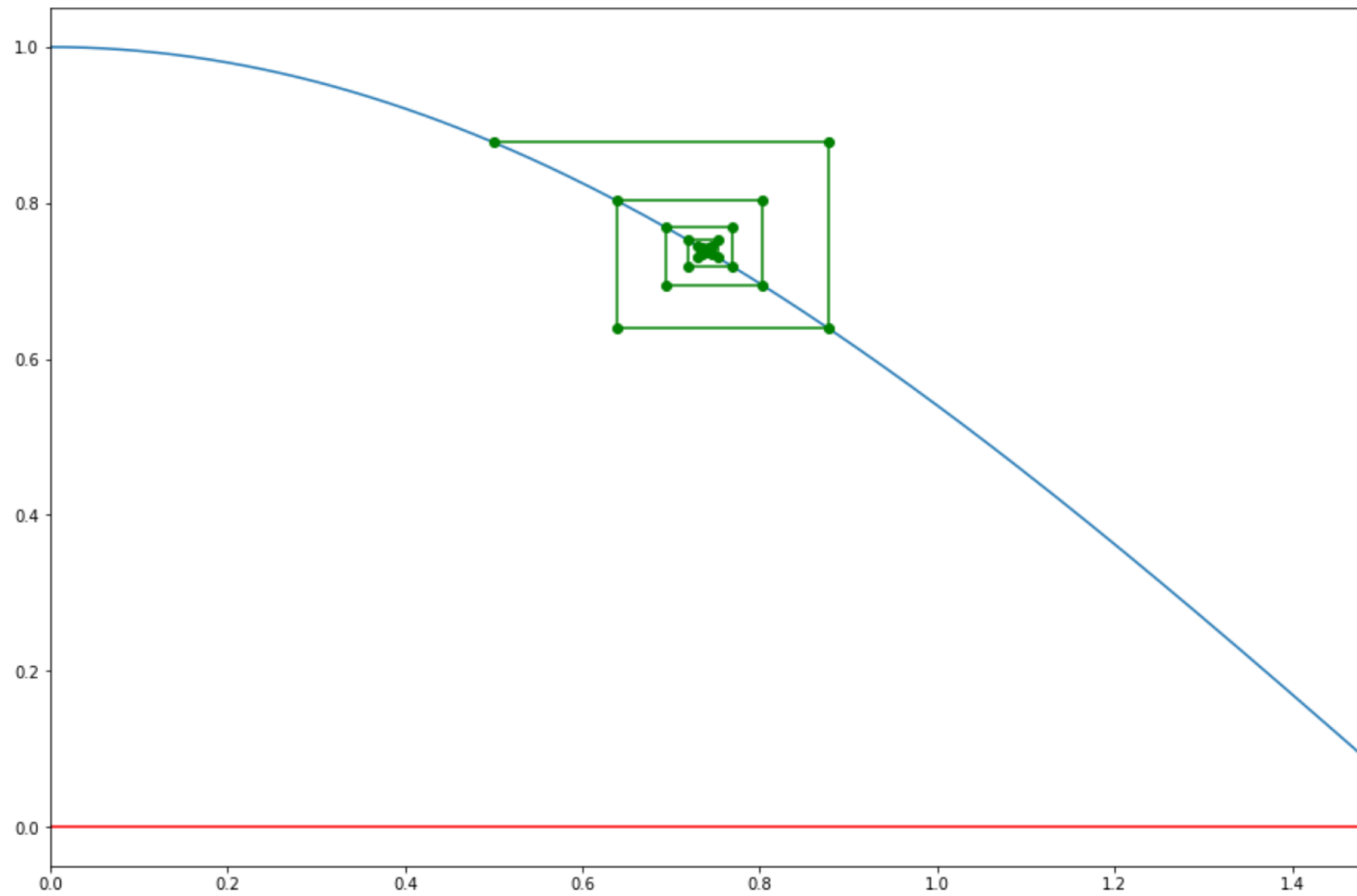
False position

False position convergence



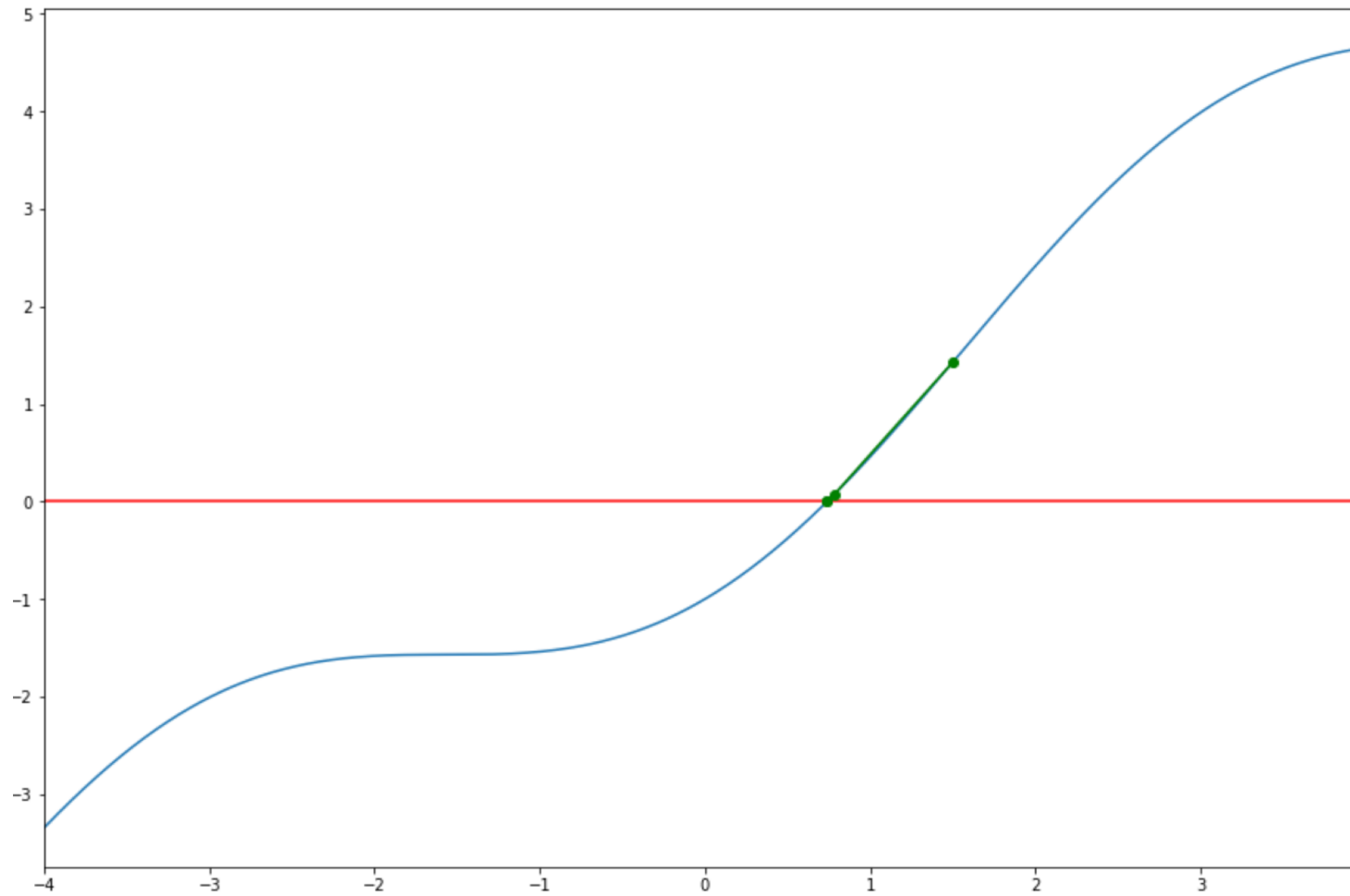
Fixed Point Method

Fixed Position Convergence



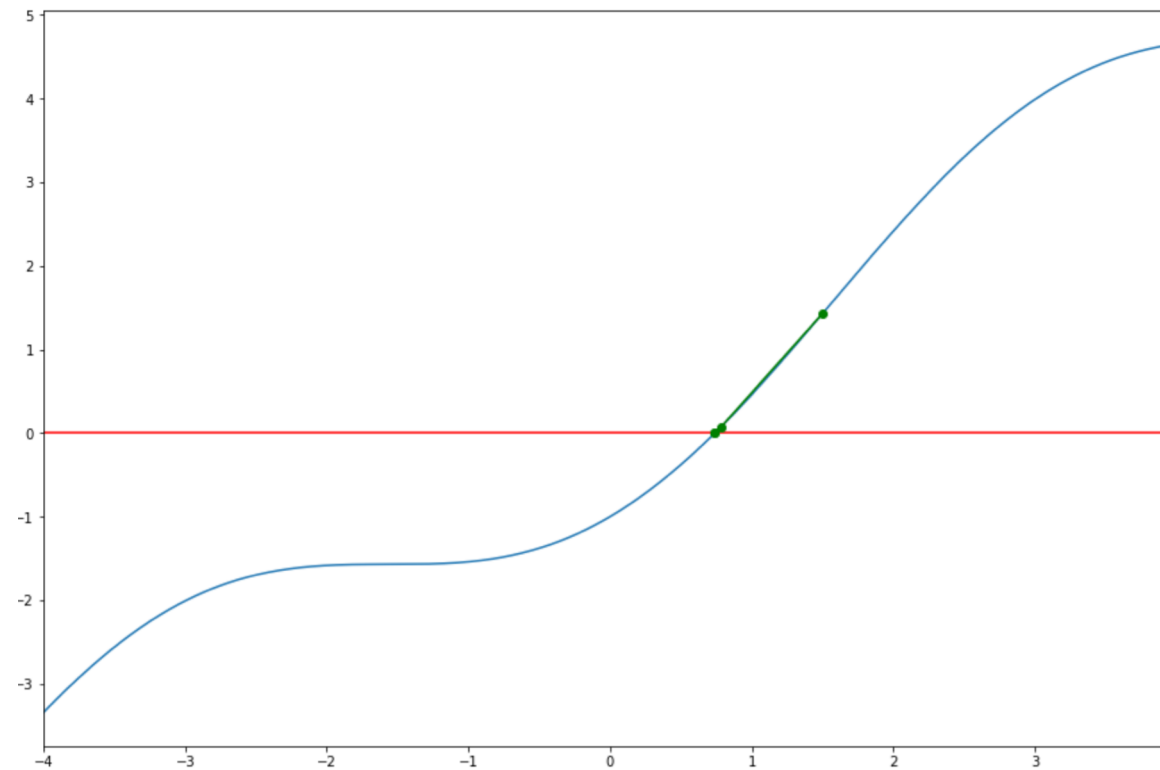
Newton Raphson

Newton Raphson



Newton Raphson Numerical Derivative

Newton Raphson Numerical Derivative Convergence

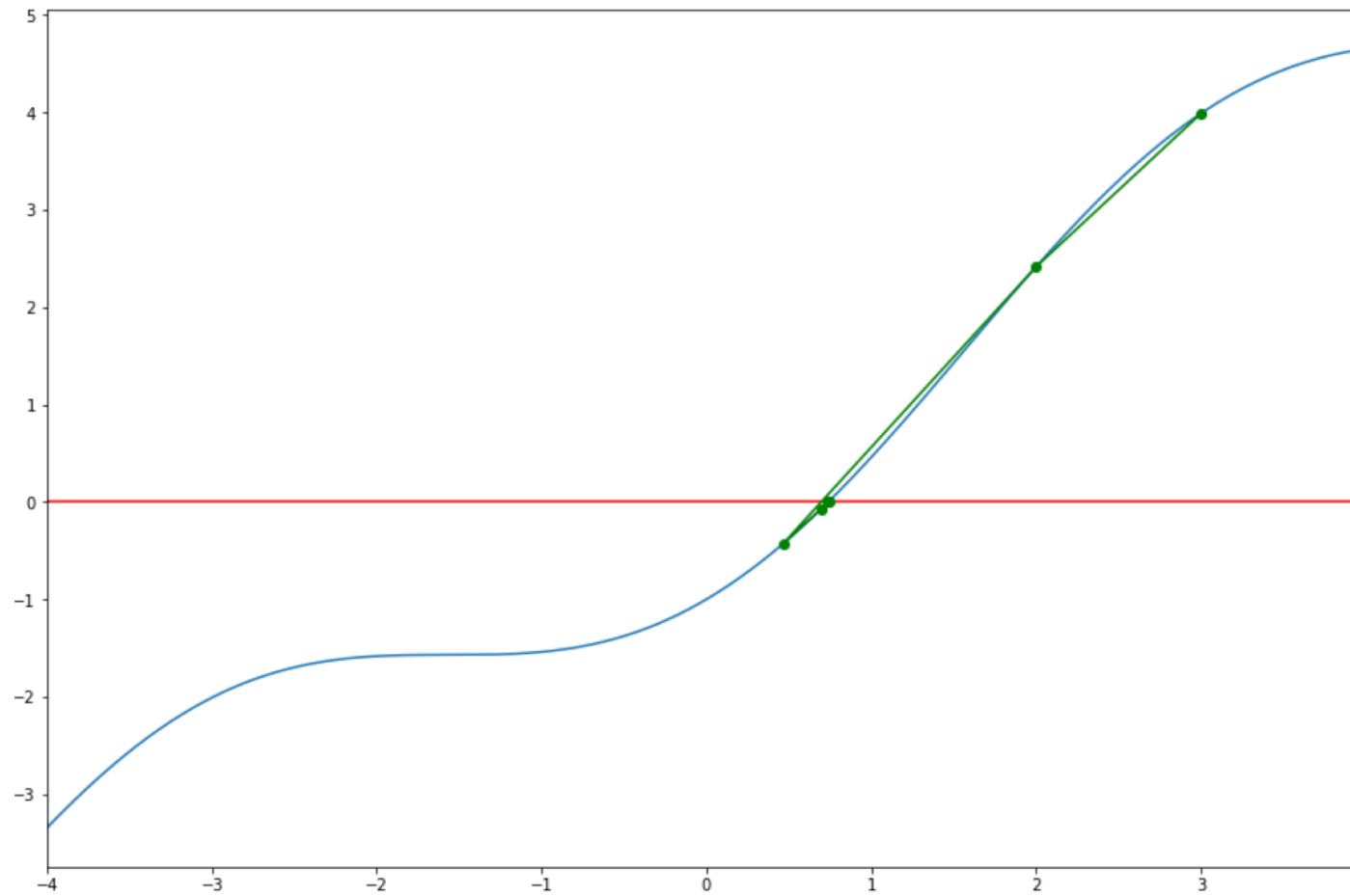


Secant

$$f'(x_k) \approx \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}$$

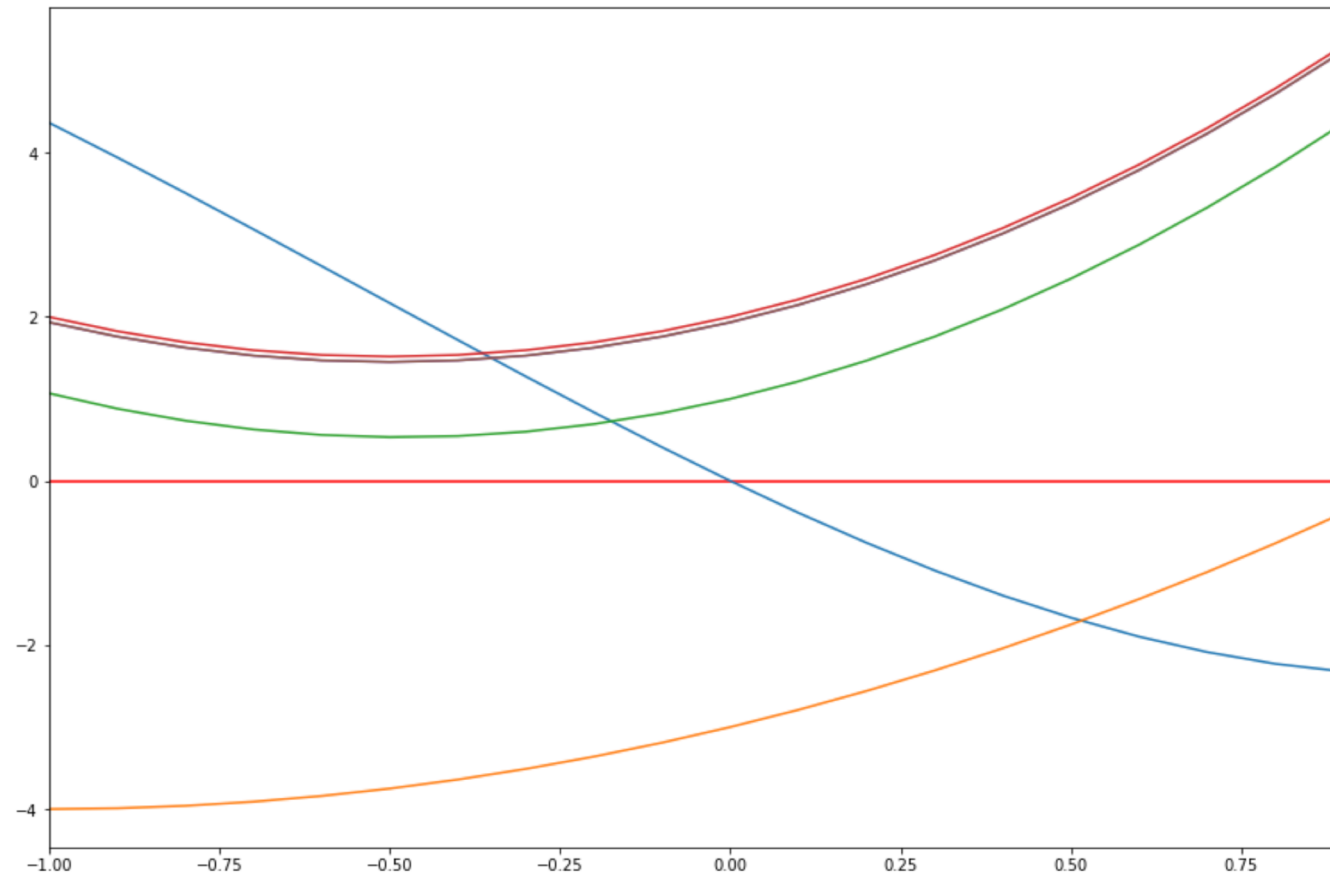
Secant

Secant Convergence



Inverse Quadratic Interpolation

Inverse Quadratic Interpolation



Linear Fraction Interpolation

5 min exercise

- Choose a function
- For that function reflect on the relationship between
 - Solver
 - Sensitivity
 - Convergence
 - Number of function evaluations

Combinations of methods

- All methods have cases where they behave poorly
- This may be addressed by mixing them
 - Choosing the best solution at any point

Dekker

Brent

Conclusions

- For something as simple as solvers in 1D there are many approaches
- Convergence rate is important for performance
 - But the number of function evaluations is also a factor
- If you know a narrow bracket the simple methods are quite efficient
 - If your function is well behaved
- Use more advanced methods if you don't know where to look
 - or if your function is ill behaved